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INSTITUTE OF ENGINEERING AND TECHNOLOGY
Approved by AICTE & Affiliated to APJ Abdul Kalam Technological University
(Owned by Adi Sankara Trust)
DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE)

AIL 411-DEEP LEARNING LAB

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EXPERIMENT 2

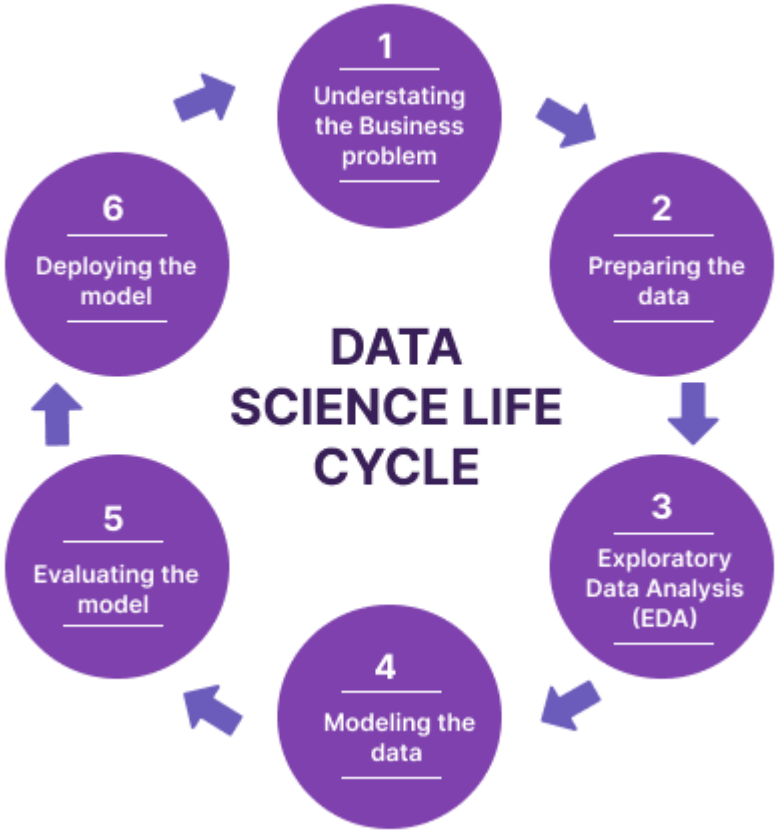
Data pre-processing operations such as outliers and/or inconsistent data value management

Agenda

- Data Science Process
- Data Pre processing
- Data Preprocessing using Titanic Dataset

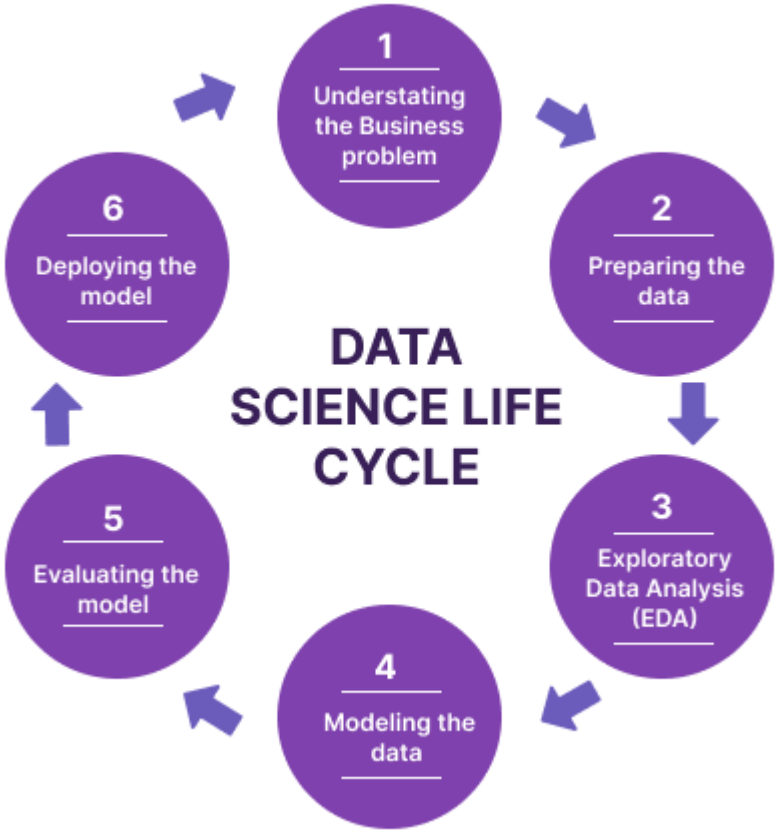
Data Science Process

- **Understand the business problem**
  - Data collection: Gathering relevant data from various sources.
- **Data preparation**: Cleaning, transforming, and organizing data for analysis.
- **Exploratory data analysis (EDA):** Summarizing and visualizing data to understand its characteristics.



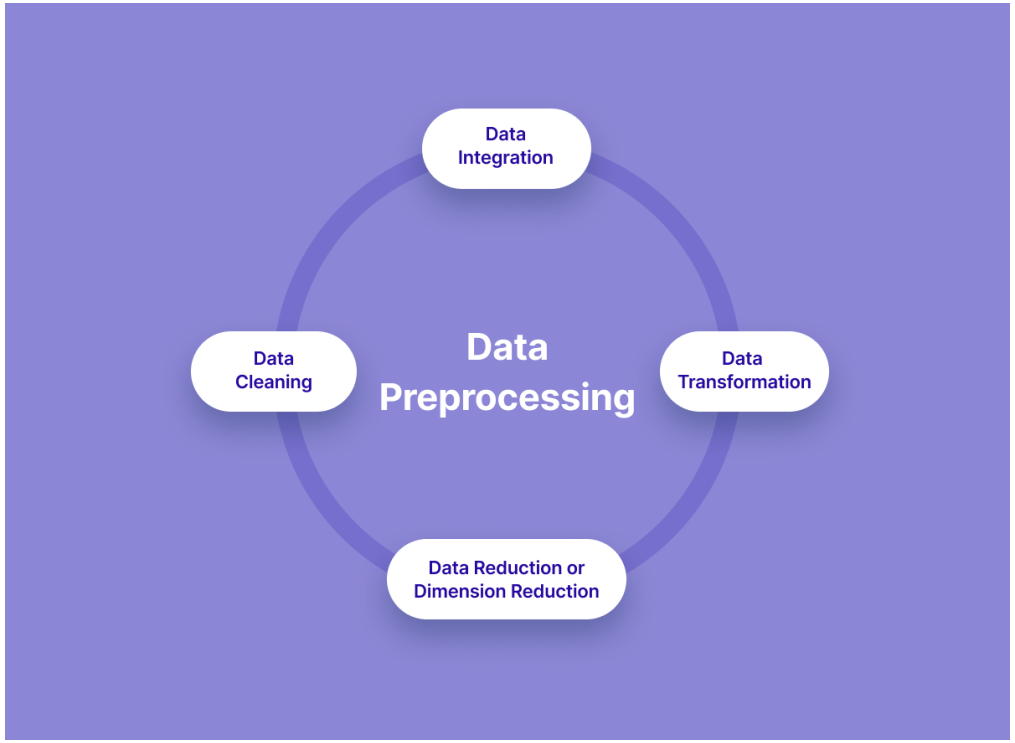
Data Science Process Cont…

- **Modeling and algorithms**: Applying statistical and machine learning techniques to build predictive models.
- **Evaluation**: Assessing model performance and refining as needed.
- **Deployment**: Integrating models into real-world applications.



Data Preprocessing

- Data Cleaning
  - Handling missing values, removing duplicates, correcting errors.
- Data Integration
  - Combining data from multiple sources, resolving inconsistencies.
- Data Transformation
  - Normalization, standardization, discretization, feature extraction.
- Data Reduction
  - Dimensionality reduction, feature selection, data compression.



Data Prepressing using Titanic Dataset

https://www.kaggle.com/datasets/yasserh/titanic-dataset?resource=download



Reference

https://www.v7labs.com/blog/data-preprocessing-guide

https://github.com/Anello92/Machine_Learning_Python/blob/main/preprocessing_machine_learning%20(1).ipynb

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https://www.kaggle.com/code/pmarcelino/comprehensive-data-exploration-with-python